H

RAFFLES INSTITUTION

2011 Year 6 Preliminary Examination
Higher 2

BIOLOGY **9648/01**

Paper 1 Multiple Choice **September** 2011

1 hour 15 minutes

Additional materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **21** printed pages.

Raffles Institution
Internal Examination

1. From which cell organelle are nucleic acids absent?

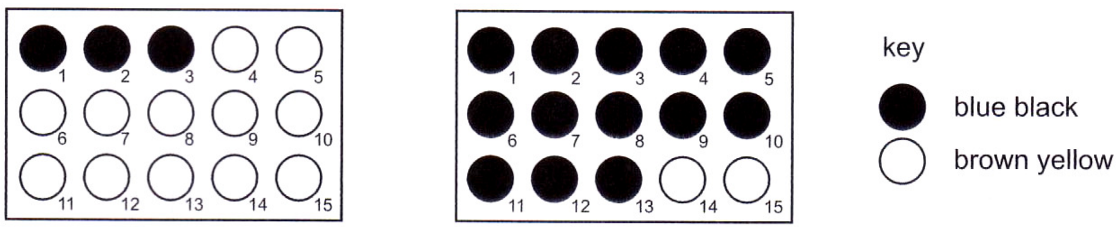
A ribosome

B chloroplast

C mitochondrion

D Golgi apparatus

2. An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and place onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

	temperature / °C	time / s
A	30 10	3.15 0.45
B	30 10	45 195
C	10 30	45 195
D	10 30	3.15 0.45

3. Which of the following best accounts for the difference in structure between amylose, which is helical, and cellulose, which is fibrous?

A Presence or absence of 180^{o} rotation of alternate subunits

B Presence or absence of branching in the macromolecule

C Subunits are either monosaccharides or amino acids

D Different tendency of macromolecule to form hydrogen bonds with water

4. At which levels of protein structure do hydrophobic interactions occur?

A quaternary only

B tertiary and quaternary

C primary, secondary and tertiary

D primary, secondary, tertiary and quaternary

5. How many different types of gametes would be produced by an organism of genotype PpqqRrSsTTUu, if all of the genes assort independently?

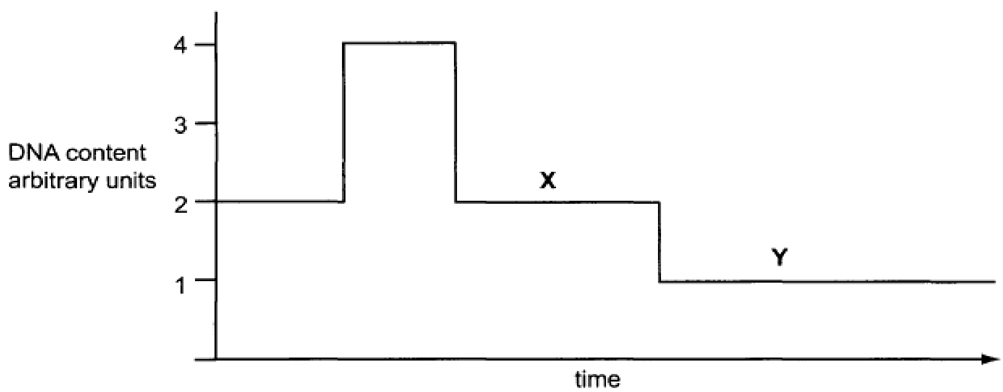
A 8

B 12

C 16

D 64

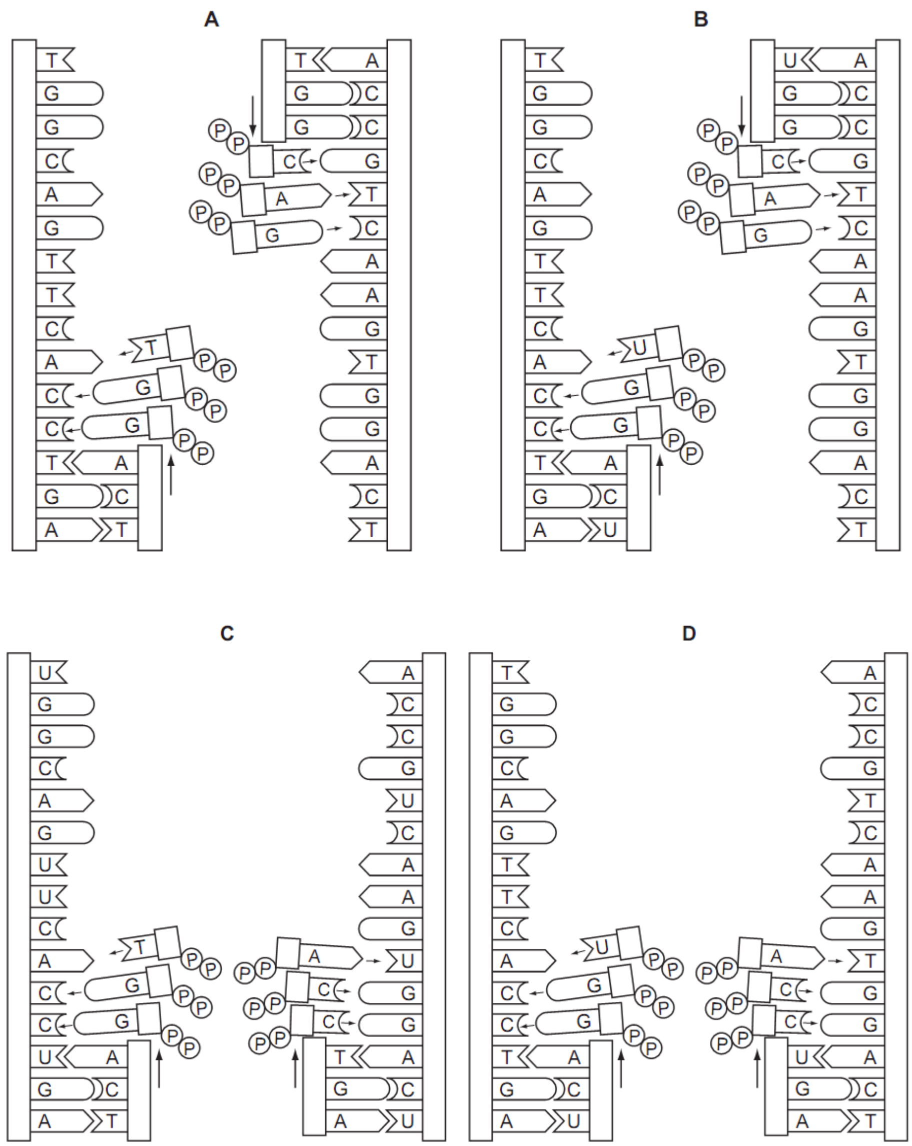
6. The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.



Which combination is correct?

	Ploidy of cell at X	**Ploidy of cell at Y**
A	diploid	diploid
B	diploid	haploid
C	haploid	haploid
D	haploid	diploid

7. Which diagram shows the semi-conservative replication of a section of a molecule of DNA?



8. In the DNA sequence for sickle cell anaemia, adenine replaces thymine in a CTT triplet, forming the triplet CAT. During synthesis of the sickle cell haemoglobin molecule, the amino acid valine is incorporated instead of glutamic acid.

What is the anticodon in the transfer RNA molecule carrying this valine?

A CAU

B CUA

C GAU

D GUA

9. The table describes different kinds of mutations in DNA.

mutation	name
from purine to other purine	transition
from pyrimidine to other pyrimidine	transition
from purine to pyrimidine	transversion
from pyrimidine to purine	transversion

The diagram represents part of a DNA molecule.

```
G A T A C C A
C T A T G G T
```

Which diagram shows the DNA molecule with only transversion(s)?

A

```
G T T A T C A
C A A T A G T
```

B

```
G A A A C A A
C T T T G T T
```

C

```
A A T A C C A
T T A T G G T
```

D

```
G A T A T C A
C T A T A G T
```

10. Some antibacterial drugs can affect the synthesis of proteins.

Antimicrobial drug	rifampicin	streptomycin	tetracycline
Mode of action	Binds to RNA polymerase	Genetic code misread during translation	Prevents binding of tRNA to ribosome

Which is the correct set of immediate effects of these drugs?

Antimicrobial drug	rifampicin	streptomycin	tetracycline
A	Defective protein synthesized	mRNA does not bind to ribosome	Amino acids not added to growing chain
B	Transcription prevented	Defective protein synthesized	Amino acids not added to growing chain
C	Defective protein synthesized	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesized	mRNA does not bind to ribosome

11.

With reference to the lambda phage shown in the diagram above, which of the following statements are **false**?

1. The genetic material of the virus can be either DNA or RNA.
2. The adsorption of viral particles to host cell is a chemical interaction in which strong covalent bonds are formed between the tail fibres of the virus and complementary receptor sites on the host cell wall.
3. Once the viral genetic material has reached the cytoplasm of the host cell, viral proteins will be degraded.
4. The host genes can be transferred by the virus to another host cell through the process of generalized transduction.
5. Mature infective virions have spontaneously assembled and are ready to leave host cell to infect other cells.
6. The virus is able to multiply via both the lytic and lysogenic cycles.

A **3**, **4** and **6** only

B **1, 2** and **5** only

C **1**, **5** and **6** only

D **2**, **3** and **4** only

12. The genome of rhabdovirus consists of a single-stranded RNA molecule whose sequence is complementary to the RNA sequence which functions as a messenger RNA. How is the (+) sequence messenger RNA produced in infected cells?

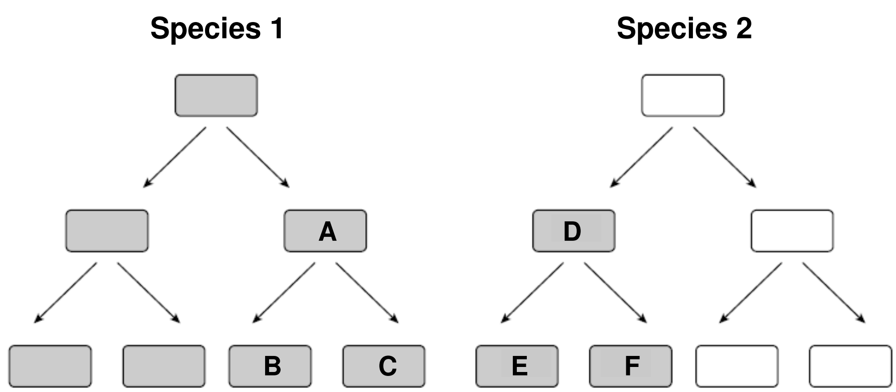
A Host cell RNA polymerase activity.

B One portion of the viral RNA genome is directly translated by host cell ribosomes.

C Reverse transcriptase activity.

D The infecting rhabdovirus contains an RNA-dependent RNA polymerase.

13. The diagram below shows 2 species of bacteria. The bacteria that are shaded are resistant to the antibiotic penicillin.



Which one of the following statements is likely to be true?

A Bacterium **D** is resistant to penicillin due the transfer of the complete F plasmid and penicillin resistant gene through transduction involving Bacterium **A**.

B Bacterium **D** is resistant to penicillin due to horizontal gene transfer involving a bacteriophage and Bacterium **A**.

C Bacterium **E** acquired resistance to penicillin as a result of random mutation and genetic transformation.

D Bacteria **E** and **F** are resistant to penicillin as a result of conjugation from Bacterium **D**.

14. Three flasks of minimal salts broth were inoculated with *E. coli*. The contents of the flasks are shown below.

Flask A
medium contains glucose

Flask B
medium contains glucose and lactose

Flask C
medium contains lactose

After a few hours of incubation, you test the flasks for the presence of β-galactosidase. What are the relative amounts of this enzyme in the three flasks?

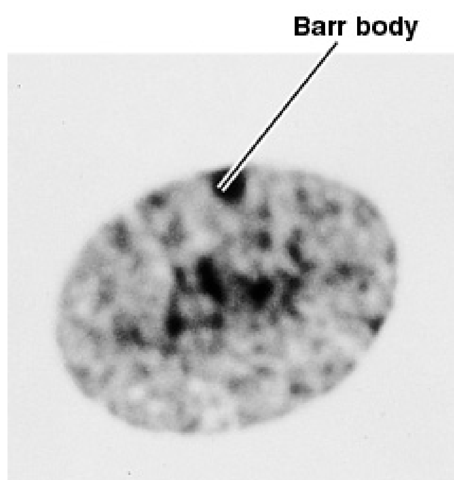
A **A > B > C**

B **B > A > C**

C **C > A > B**

D **C > B > A**

15. The figure at right shows a Barr body in cells of a normal human female. Which of the following statements about X chromosome inactivation is **false**?



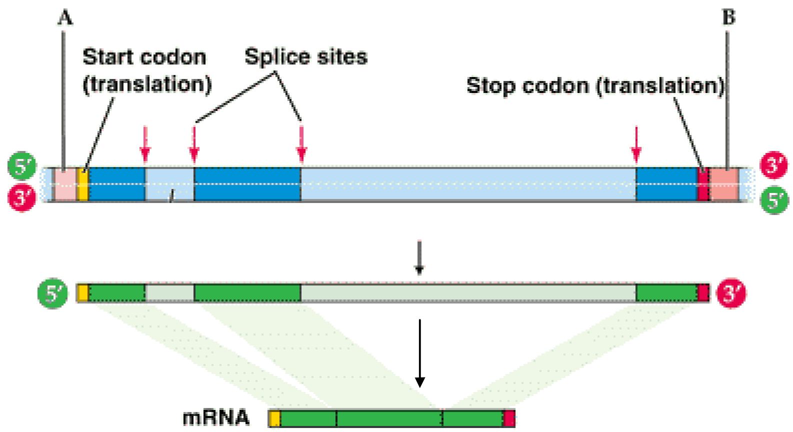
A The probability that a given X chromosome will become inactive is 50%.

B Inactivation of the X chromosome seems to involve methylation of cytosine on DNA.

C The Barr body is a clump of euchromatin representing the inactivated X chromosome.

D An abnormal male with the genotype XXY would have one Barr body.

16. In the figure shown below, A is the _______ region, B is the _______ region, and this gene includes _______ introns and _______ exons.



A promoter; terminator; 2; 3

B promoter; terminator; 3; 2

C terminator; promoter; 2; 3

D terminator; promoter; 3; 2

17. A student has 2 cultures of cells. Culture A consists of normal cells, while culture B consists of cells which were isolated from a tumor and divide uncontrollably. He wants to determine if the uncontrolled growth of cells in culture B was the result of activation of an oncogene or the mutation of a tumour suppressor gene. He decides to fuse the cells from the two cultures to form a hybrid line containing DNA from both cells.

The list of possible results and the conclusions drawn from the experiment are summarized by the student in the table below:

	Resulting growth of hybrid cells	**Type of gene mutated in the tumor**
(1)	Grows normally	Activated oncogene
(2)	Grows uncontrollably	Activated oncogene
(3)	Grows normally	Mutated tumour suppressor gene
(4)	Grows uncontrollably	Mutated tumour suppressor gene

Only two of the conclusions are correctly matched with the possible results. The two correct possibilities are

A (1) and (3)

B (1) and (4)

C (2) and (3)

D (2) and (4)

18. A man and a woman, both with normal colour vision, have a colour-blind boy together. The woman is pregnant for a second time, and the doctor tells her she is carrying twins of one boy and one girl. What is the chance that both twins will have normal colour vision?

A 0

B ¼

C ½

D ¾

19. A man was brought to court in a paternity case. He has blood type B, Rh positive. The mother has blood type B, Rh negative.

What blood type of the child would exclude the accused from paternity?

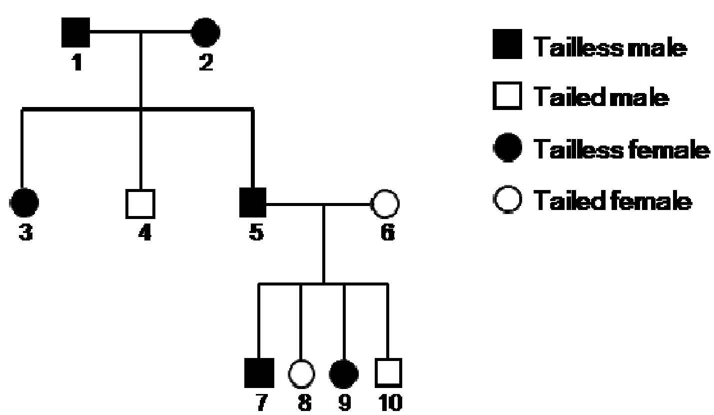
A AB, Rh negative

B B, Rh negative

C B, Rh positive

D O, Rh negative

20. The Manx gene determines the presence or absence of tails in cats. Individuals that do not possess the dominant allele develop tails. The pedigree chart below shows the absence or presence of tails in three generations of cats.



Which one of the following statements is likely to be true?

A Individual 2 is homozygous dominant for the Manx gene.

B If individual 4 mated with another tailed cat, they would produce tailless offspring.

C The homozygous dominant condition is lethal.

D The Manx gene is located on the sex chromosome.

21. Aberrations during crossing over in Prophase I of meiosis may result in chromosomal mutations. Which of the following mutations is least likely to result?

A Deletion

B Duplication

C Polyploidy

D Translocation

22. Horses are capable of producing two basic pigments: black and red pigments. The recessive **e** allele allows a horse to produce only **red pigment** in the hairs of its coat. The dominant **E** allele allows a horse to produce **black pigment** *as well as* red pigment in the hairs of its coat.

The Agouti (**A**) locus controls *where* on the horse the **black** pigment shows up. The alleles involved here are a dominant **A** and a recessive **a**. The dominant **A** allele causes the black pigment to be restricted to the horse's **extremities** which include lower legs, mane, tail, muzzle and eartips.

Horses may be **Bay, Black** and **Chestnut**. A horse that is chestnut (**ee**) has no black pigment. A Bay horse has a red body with black extremities.

If two black horses are mated, which of the following coat colours will definitely **not** show up in their offspring?

A Bay

B Black

C Chestnut

D All three colours are possible

23. Which one of the following would be expected to increase the amount of ATP produced through operation of the respiratory chain and chemiosmosis?

A Increasing the permeability of the inner mitochondrial membrane to protons (H^+)

B Decreasing the pH in the space between the inner and outer mitochondrial membranes

C Increasing the H^+ concentration in the mitochondrial matrix

D Reducing the supply of NADH + H^+

24. Consider two groups of isolated thylakoids which have been placed into an acidic bathing solution so that H^+ diffuses into the thylakoids. If they are then transferred to a basic bathing solution with one group placed in the light with the other group kept in the dark, select the choice that describes what would be expected.

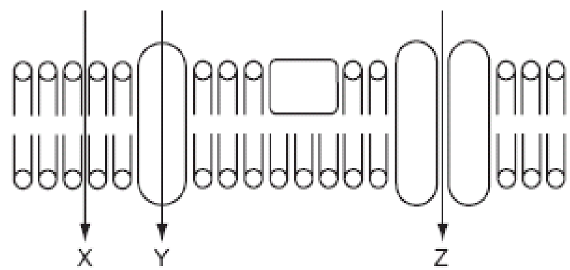
A Light: ATP only; Dark: Nothing

B Light: ATP, O_2; Dark: ATP only

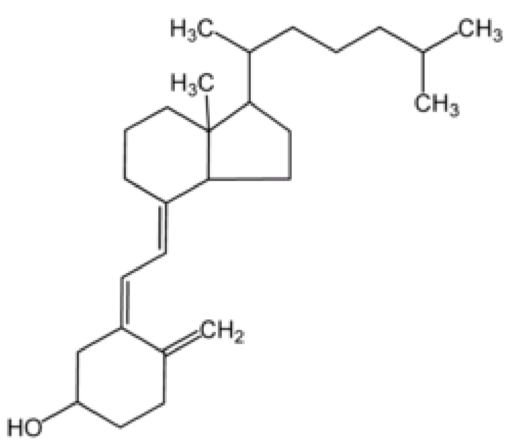
C Light: ATP, O_2, glucose; Dark: ATP, O_2

D Light: ATP, O_2; Dark: O_2

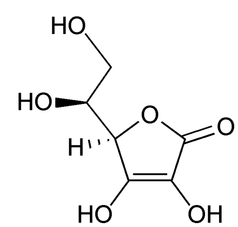
25. The diagram shows three routes through which substances can pass across a cell membrane.



Which correctly shows the routes for molecule **1** and molecule **2**?



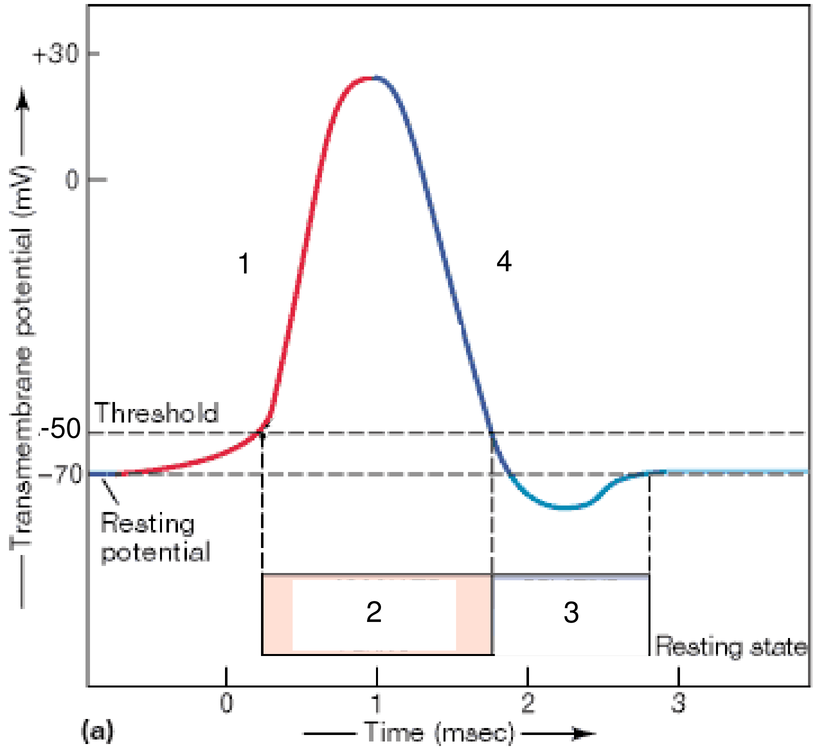
molecule **1**



molecule **2**

	molecule **1**	molecule **2**
A	Y	X
B	X	Z
C	X	Y
D	Z	Y

26.

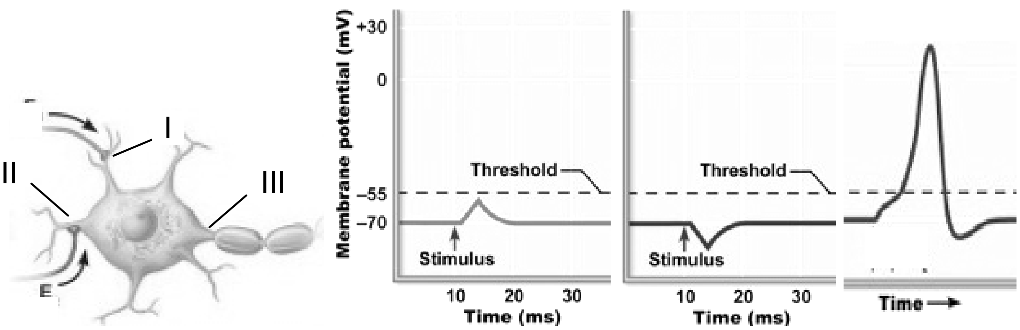


Identify the stages 1-4 in the diagram above.

	1	2	3	4
A	Depolarisation	Absolute refractory period	Relative refractory period	Repolarisation
B	Depolarisation	Action potential	Hyperpolarisation	Repolarisation
C	Rising phase	Relative refractory period	Absolute refractory period	Decreasing phase
D	Repolarisation	Absolute refractory period	Relative refractory priod	Depolarisation

27. A neuron receives inputs from 2 other neurons as illustrated in the diagram below.

The graphs show the membrane potential at points I, II and III on the post-synaptic neuron.



I II III

If the voltage-gated calcium channels on the pre-synaptic membrane at point **II** are no longer able to respond to a depolarisation, what is the expected effect on the post-synaptic neuron?

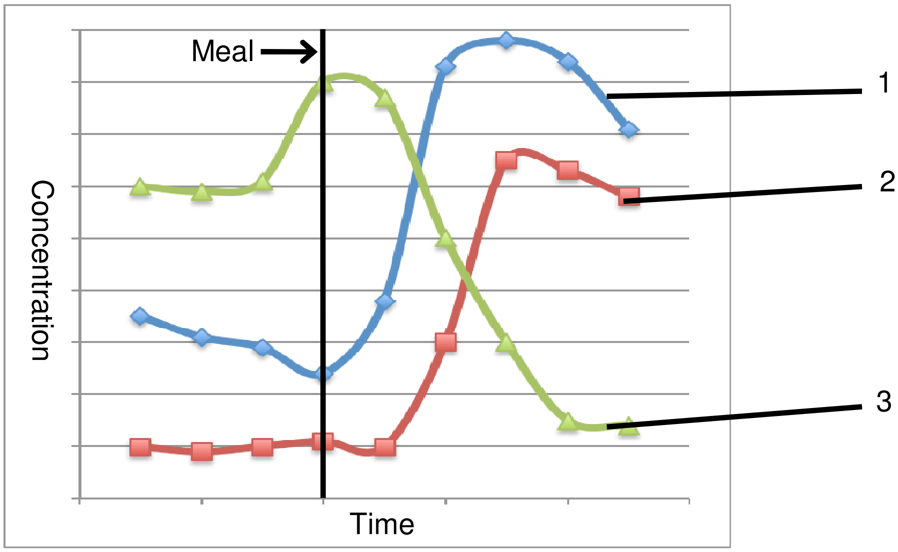
A The post-synaptic neuron is now unable to receive any inputs.

B Post-synaptic neuron is less depolarised and a stronger than normal stimulus is required to initiate an action potential in the post-synaptic neuron.

C Post-synaptic neuron is more depolarised and a weaker than normal stimulus is sufficient to initiate an action potential in the post-synaptic neuron.

D Post-synaptic neuron is hyperpolarised and a weaker than normal stimulus is sufficient to initiate an action potential in the post-synaptic neuron.

28. The graphs below show the changes in the concentrations of glucose, insulin and glucagon in the blood after a meal.



Identify what graphs 1-3 correspond to.

	1	2	3
A	Glucose	Insulin	Glucagon
B	Insulin	Glucose	Glucagon
C	Glucagon	Insulin	Glucose
D	Glucose	Glucagon	Insulin

29. Which of the following statements are true?

1. All receptor tyrosine kinases comprise 2 subunits which come together upon ligand binding.
2. Receptor tyrosine kinases are able to phosphorylate tyrosine residues in relay proteins.
3. G-protein coupled receptors have 7-transmembrane regions.
4. Hydrophobic ligand molecules typically bind to intracellular receptors.

A 1 and 4

B 2 and 4

C 2, 3 and 4

D All the above statements are true.

Use the following information for Questions 30 and 31.

The Norway rat (*Rattus norvegicus*), a widespread pest, was controlled for about a decade by the anticoagulant warfarin. This chemical substance inhibits the clotting of blood. Resistance to warfarin is governed by a dominant autosomal gene, *R*. The table below indicates the response of individuals to warfarin, vitamin K dependence and relative reproductive fitness. The highest fitness level is 1.00.

Characteristics	**Genotypes of individuals**		
	rr	***Rr***	***RR***
response to warfarin	susceptible	resistant	resistant
vitamin K dependence	no	intermediate	high
relative fitness	0.68	1.00	0.37

30. Which of the following is correct concerning the gene for resistance to warfarin?

A It is a wild-type gene induced to mutate by warfarin.

B It is a mutation favoured by natural selection.

C It is increased in frequency by virtue of its dominant status.

D It offers an advantage to fitness levels under all conditions.

31. In a population where warfarin is no longer applied, the expectation is that the

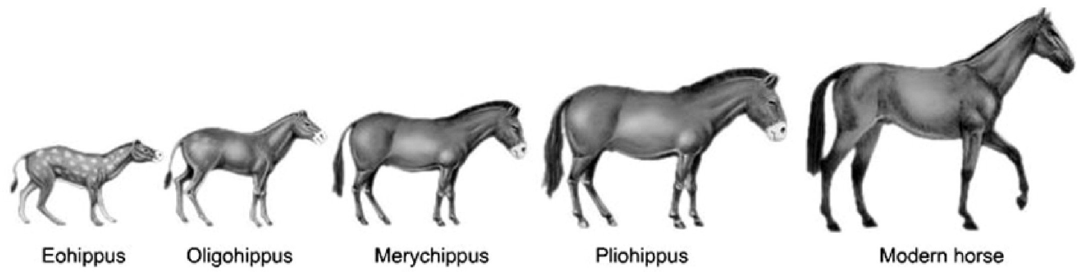
A heterozygotes will have a reproductive advantage over both homozygotes.

B frequency of the dominant allele will decline.

C dependence of the population on vitamin K will increase.

D proportion of recessive individuals will gradually decline.

32. The diagram below suggests the evolution of horses beginning from the Eohippus 58 million years ago.



Fossil records suggest that the ancestor of the modern horse is believed to have had relatively short legs. According to Darwinian views, the evolution of long-legged horses is best explained by

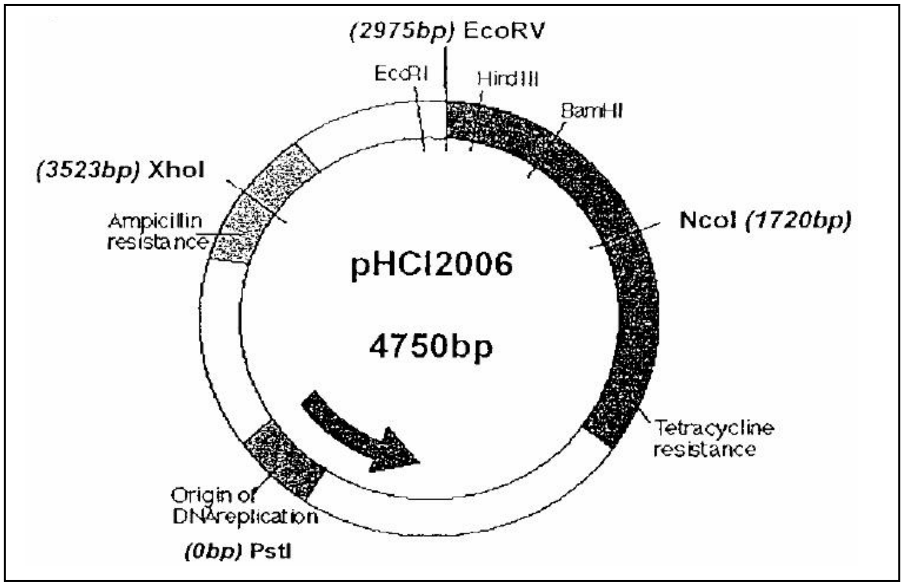
A stabilising selection.

B directional selection.

C disruptive selection.

D acquired characteristics.

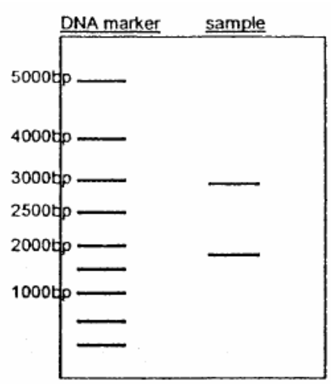
33. A gene Y, of size 726bp, is inserted into the vector, *pHCl*2006, using *XhoI*.



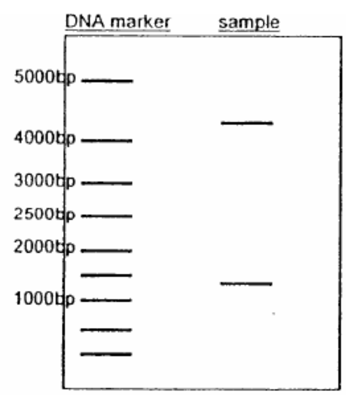
The recombinant DNA is then digested with ***Eco*RV and *Pst*I** and separated by size through gel electrophoresis. The sizes of the separated strands are determined by comparison to a DNA marker.

Which of the following gel electrophoresis diagrams shows the correct sizes of the separated strands?

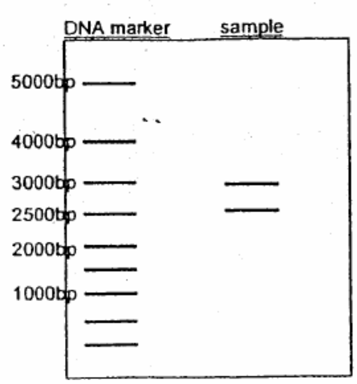
A



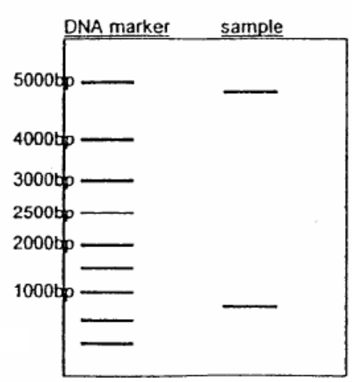
C



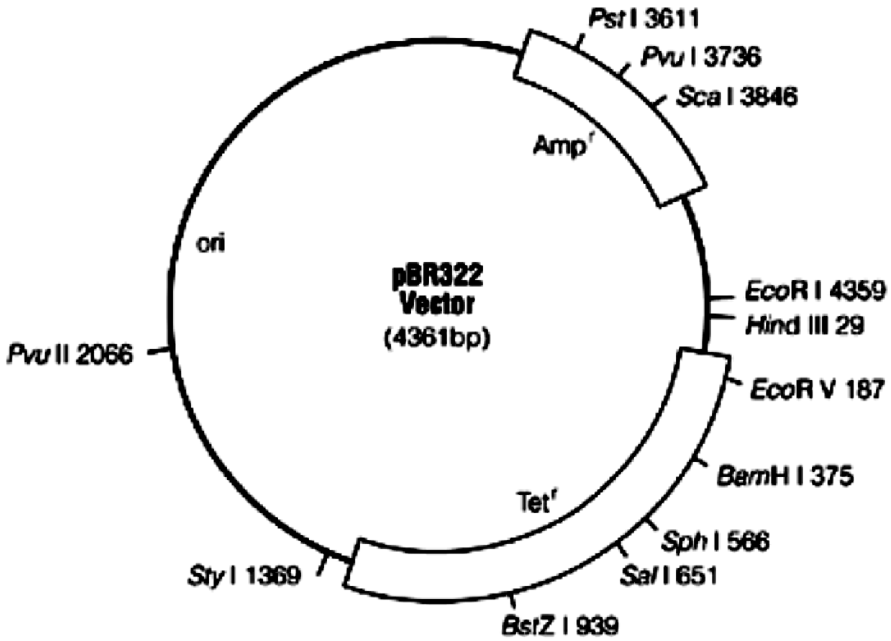
B



D



34. The pBR322 vector is used to clone a eukaryotic gene, which has been digested by the restriction endonuclease *Bam*HI.



Following transformation, bacterial cells were grown in four different media, as shown below:

I nutrient broth plus ampicillin

II nutrient broth plus tetracycline

III nutrient broth plus ampicillin and tetracycline

IV nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grow in?

A **IV** only

B **I** and **II** only

C **II** and **III** only

D **I** and **IV** only

35. A student performed an experiment to determine if a particular gene has been inserted in a genetically modified organism.

i. Transfer of DNA to nitrocellulose membrane

ii. Restriction digestion of genomic DNA

iii. Cleaved DNA separated using gel electrophoresis

iv. Create radioactive probe

v. Incubate probe and membrane

Which is the correct sequence to the above steps?

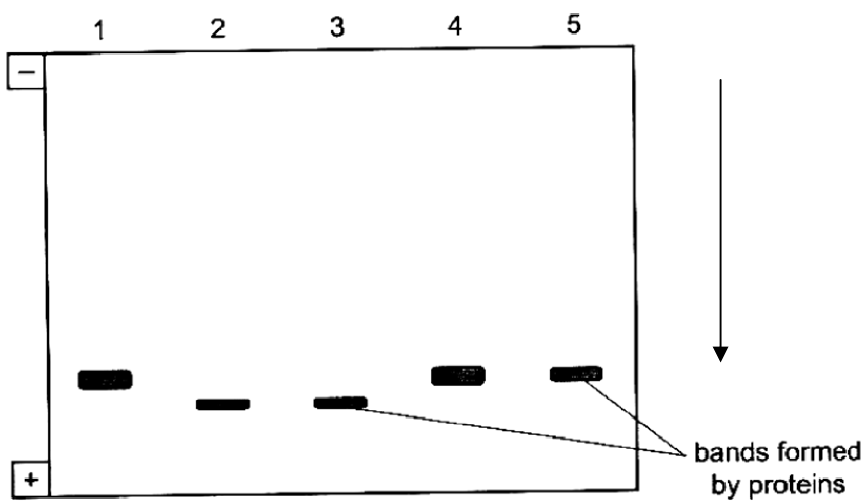
A ii → iii → i → v → iv.

B ii → iii → i → iv → v.

C iv → v → i → ii → iii.

D iii → i → ii → iv → v.

36. The diagram shows the results of electrophoresis of the same protein extracted from five different individuals.



A mutation occurred in individuals 2 and 3. Which one of the following can account for the different banding pattern?

A A base has been deleted.

B A base has been added.

C A negatively charged amino acid replaced a non-charged amino acid.

D A positively charged amino acid has been added to the protein.

37. Which of the following is a suitable source to locate stem cells?

A Human blastocyst

B Umbilical cord blood

C Liver

D All of the above

38. Why are somatic line therapies licensed for the treatment of genetic disorders rather than germ line therapies?

A Somatic cell manipulation is more effective than germ cell manipulation.

B There is no control where the DNA inserts in germ line therapies.

C The transformation is not carried on to the next generation in somatic line therapies.

D Viruses can be used as vectors in somatic line therapies.

39. Which of the following is/are reasons for scientists to employ the method of plant cloning?

1 to produce more of a selected phenotype of an organism

2 create smaller genetic changes at a much more rapid pace

3 production of large amount of pharmaceuticals

4 selective breeding is too slow

5 to prevent natural selection from occurring

A 1,2,3 and 4 only

B 3 only

C 2,3, and 5 only

D 1,3,4 and 5 only

40. Large quantities of useful products can be produced through genetic engineering involving

1 bacteria containing recombinant plasmids

2 mammals producing substances in their milk

3 transgenic plants

4 microorganisms producing human hormones

5 determination of number of genes in human

A 1,2,3,4 only

B 2,3,4 only

C 1,3,5 only

D 2,4,5 only